

04/30/2026

1. Clean up the slides + short summary report before thursday sent to Justin.
2. After the chains and springs arrive.
 - a. [Manav] close the chain loop and adjust the tension/sprocket distance
 - i. Better friction fit for bearing with added groove for flange
 - b. [Leah] reprint the divider
 - c. Assemble the whole system
 - i. do similar tests as before
 1. Static test → change the surface incline and report the failure point
 2. Dynamic testing → find a suitable incline to drive the system and investigate the valve switching performance/leakage
 - ii. You need to lock the motor in code or use dynamixel wizard to prevent rotation
 - iii. Get information about the locaiton oft flipping magnets polarity.

04/23/2026

Orders:

- 1 more chain
- Springs

Write the report: as the semester is ending, please start to organize the documents and prepare the end-of-the-semester report. As Justin said, you can use a similar literature review as before, and not too much detail in the report, as he mostly joined the meetings, yet make sure you include all the design steps, tests, results, etc.

Leah:

1. CAD + print the port to divide the vacuum pump into two driving plates
 - a. Make sure you consider how you want to attach it, either to the pump itself, the base, or any other way. It is better to be connected to something, so the tube is better organized.
2. Cut the vent screws to the correct lengt
3. Test the pneumatic valve with different softnesses of silicone O-rings for self-sealing.
 - a. Ratio 15:1 up to 20:1
4. Suspension system
 - a. Assemble the spring to the system
 - b. [Future prototype] Attach the springs between the driving plate and the gelbelt side plate
 - c. [Future prototype] Fix the tolerance between the beam and the top plate

Manav:

1. Last week's results
 - What was the weight and rough center of mass height in the previous test setup?
 - For the suction cup contact test, did you increase or decrease the height of the system after you used different sprockets? The distance should be adjusted relative to the chain location. Accordingly, the larger sprocket should have better contact, but your results show a different conclusion. Was the small sprocket aligned correctly or the larger one?
2. Update the CAD design
 - Remove the top sprocket row. We only need the bottom one, unless the top and bottom chains are too close to have the pneumatic junction in between. Yet, in that case, you can just use a slightly bigger sprocket.

- Consider the motor attachment to the plate and the coupling of the sprocket to the motor.
- Add some alignment grooves/lines on the side plates that show where exactly the magnets should be attached to the surface.
- Print the side plates and the sprockets after updating the design.

3. Assemble the chain on the printed structure and address any issues you encounter.

Shared tasks (you can work together or divide based on your preference)

1. Assemble the side plate system, the middle structure, and the gelbelt.
2. Assemble the new valves and suction cups to the chain using the newly arrived chain links.
3. Drive the motors to see whether the chain structure functions well.
4. If the second chain and spring arrive, you can assemble both sides.

04/16/2026

Leah:

5. Update the pneumatic valve design
 - Consider the vent screws
 - Consider putting the valve next to the chain attachment along the chain path
 - Make sure the design does not interfere with the sprocket
 - Fabrication and test attachment on the chain
6. Compression spring design considerations.
7. Cut the vent screws to the correct length
8. Assemble the suction cups, valves, chain, and other pneumatic system components

Manav:

4. Do the previous test again with weights attached closer to the contact plane.
 - Test in different angles (from horizontal to vertical and two in between)
 - Hold for 1 min
 - Consider 15 N for the entire test setup (include chain, pneumatic system, etc.)
5. Test the suction cups to get in contact
 - Print a mount to attach the suction cups to the chain
 - Print and assemble the sprocket/chain/suction cups
 - Test the in-contact folding by manually rotating the sprockets.
6. After Leah has the updated valve design → update the side plate
 - Consider the magnet location based on the new valve
 - Have the motor on the inner side of the plate (you should possibly be able to connect directly to the sprocket axis?)

04/09/2026

Manav;

- ☐ Put the chain to valve attachment piece on the purchase sheet.
- ☐ Simulate the landing angle of the suction cups with different pulley sizes and configurations (check 2 and 4 configurations and how it improves).
- ☐ Side plate initial design. Use compatible pulleys with the chain.

Leah:

- ☒ ~~Order the magnets for the valve system~~
- ☐ Update the design of the base, etc.
 - ☐ Beams attach flat to the GelBelt plate.

- ☐ Print the components and assemble them.
- ☒ ~~Update the valve design to be attachable to the chain.~~
- ☐ Print the new valves and test.
- ☐ Explore the new compression system.

03/12/2026

Leah

- ☒ ~~1-minute attachment test with 6 valves (video)~~
- ☐ Junction testing
- ☐ Hand rotation of the tubes with the new junction

Manav

- ☐ Pull test with suction cups attached to the chain
- ☐ The Modeling:
 - ☐ The chain is rigid, but the angle of the chain to the wall changes, resulting in nonuniform forces being applied to the suction cups.
 - ☐ Input for the model will be the angle of the wall, the stiffness of the suction cups, and the center of mass related to the suction cups.

02/19/2026

Leah

- ☐ Test the new junction
- ☐ Order the new magnets

Manav

- ☐ CAD the test setup
- ☐ Fabricate the test setup

02/12/2026

Order:

- ☐ Chain and the guide (one of each)
- ☐ Threaded screw and other connection parts

Test:

- ☐ Design the test system for the chain and guide
 - Forces and torques should be tested in different directions (pull down, pull sideways,

torsion)

- ☐ Assemble the belt and tubes to check the rotary joint resistance

02/05/2026

Sprocket and roller pulley chain

- ☐ Calculate torque modeling
- ☐ Component selection for sprocket, belt, and attachments

Pneumatic Components

- ☐ Design a bearing component for the connection between the rotary joint and the junction
- ☐ Find the threaded pipe for the connection between the suction cup and the pneumatic valve

11/17/2025

IROS:

Design

- ☒ ~~Attach to the surface~~
- ☐ Two-sided drive (two motors)
- ☐ Valve with small holes for an air vent

Experiments

- ☐ Roll slowly and slightly turn
- ☐ Go around the cylinder

11/17/2025

Order list:

- ☒ ~~DC motor for drive~~
- ☒ Rotary Joint
- ☒ Magnets
- ☐ Vacuum Valve spring
- ☒ ~~DC motor for suspension~~
- ☐ Possibly solenoid valves
- ☒ Pulleys
- ☒ Belt
- ☒ ~~Lead screw~~
- ☐ Bearings
- ☐ Partially threaded screw

11/10/2025

- Curved surface
 - How do we press into the surface as we move forward (individual cup suspension or loosening belt)
- Motor orientation
- Timing belt (size, length, cut)
- Attaching the joint to the belt has to be flexible
- Magnetic strip
- Gluing a loop to the suction cups for wire detachment

1. Active Suspension

- a. Motor attached to a spring (better to have some worm gears in the gearbox)
 - i. <https://www.robotshop.com/products/e-s-motor-12v-micro-dc-worm-gear-motor-16rpm?qd=c333400f06494a9ace8e1d25cfa68521>

- ☐ Check available designs for active suspension
- ☐ Concept design of the suspension in the robot
- ☐ Model the active suspension (displacement to normal force)
- ☐ Choose the motor and the spring

2. Pneumatic subsystem

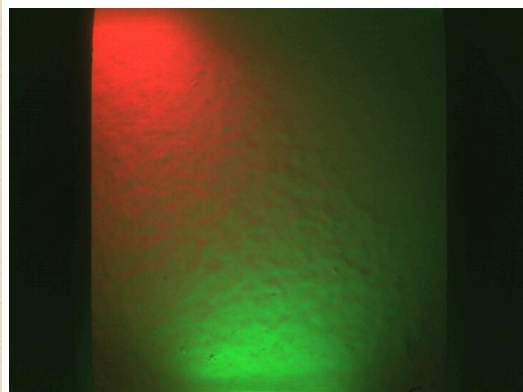
- a. Suction cup switching (activate and deactivate)
 - i. Magnetic valve
- b. Rotary mechanisms for the tubes
 - ☐ It will be good to quickly check if there are better alternatives than our plan
 - ☐ CAD the concept and try to print a rapid prototype to check the feasibility (we have steel balls, magnets, and some springs in RoboTouch)

3. Belt

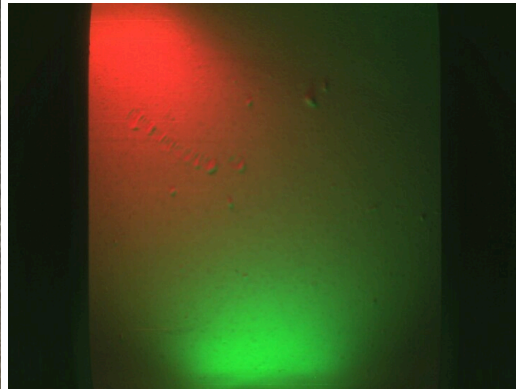
- a. Suction cup structure attachment to the belt
- b. Concept design of the locomotion in the robot system

11/03/2025

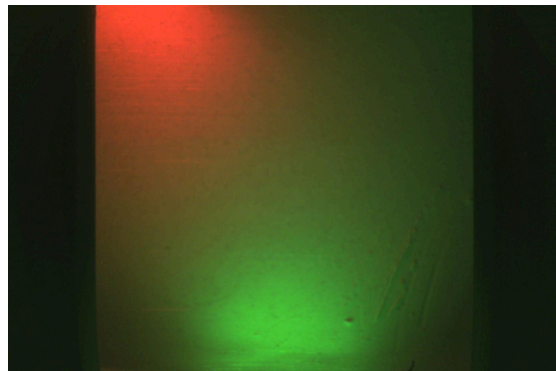
- Normal test setup printed
- Assemble test setup
- Test the setup
 - ☒ ~~Inverted and tilted surfaces (acrylic and the aluminum plate)~~
 - ☐ Three springs
 - ☐ Offsets (between the sensor and the suction cups)
- Evaluation criteria
 - ☒ ~~Failure (fall)~~
 - ☒ ~~Quality of image~~
- Result analysis and share it with PIs
 - Medium spring (5.6N/cm)
 - 2115grams



Medium spring (5N/cm) on wall, vertical



- Medium spring (5N/cm) on aluminum, upside down



- Small spring (3N/cm) on acrylic, upside down (not clear)

10/20/2025

- ☐ System modeling (forces and deformations)
- ☐ Spring selection
- ☐ Finalize the CAD for the normal test (make it assemblable to the gelbelt)
- ☐ Start printing and assembling after the meeting and the suction cups' arrival

10/09/2025

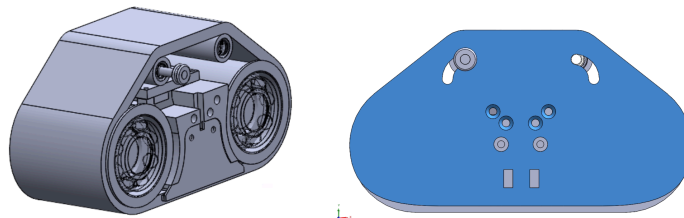
- Ordering suction cups
- Pneumatic switch
- Normal Force Setup:
 - ☐ Suspension
 - ☐ Vacuuming system
- Next:
 - ☐ Belt and pulley system
 - ☐ Drive the wheels

10/09/2025

Evaluation:

- Total Force: $\sim 15 + 10 = 25 \text{ N}$
 - GelBelt: $\sim 13 \text{ N}$
 - Weight: 3 N
 - Press: 10 N
 - Leah: $\sim 10 \text{ N}$ for 4 motors
 - Motor and the individual linkage: 2 N
 - Base: 2 N
 - Manav: $\sim 10 \text{ N}$
 - Suction cup and tube: 1 N
 - Base and mechanisms: 5 N
- How many actuators do we need?
 - Manav: $\text{Max}(25/F_N, 15/F_T) \rightarrow$ getting 6 blue suction cups for now?
 - Leah: $25/F_N \rightarrow$ We might need bigger motors
- Power consumption:
 - $V \cdot I$
- Steering:
 - Think about the steering mechanism for both the inverted surface and the wall.
 - GelBelt should be disengaged
 - The motion is better to happen close to the surface
 - For the suction cup, we need a separate arm to attach and rotate
 - For the quadcopter, we might need small adjusting legs to maintain contact with the surface while the controller rotates the quadcopter.
- Performance on the curved surfaces

Attach to Gelbelt for normal force test:



10/03/2025

Next Week

- Manav:
 - Fix the mount for all the suction cups and do the testing for F_z and F_t with vacuuming.
 - Plot the normal and tangential results in a 2D plot so that we can get the failure boundaries
 - We should think about surfaces with different roughness.
 - Hopefully, we can get an aluminum with different surface finishes
- Leah:
 - We either need bigger motors or a larger number of motors.
 - Right now, each motor $\sim 3\text{N} \rightarrow 4\text{motors } 12\text{ N} - 2\text{N motor weight} = 10\text{ N}$
 - We might be able to make the propeller closer to the surface.
 - Test on the curved surfaces.
 - Plot the data

Next Goals (After the test stage):

- How many suction cups/motors do we need?
- First prototype for normal force:
 - Test only normal force with the gelbelt to see if we get good readings.
 - The suction cup will need a suspension/push mechanism to press the GelBelt to the surface.
- Evaluation:
 - Power
 - Steering
 - Weight and force/weight
 - Performance on the curved surfaces (aircraft plane)
- We check to see if one method is better than the other
 - We continue toward the detailed design and add the movement on the surface

09/19/2025

******* Finalizing Orders**

Leah:

- ~~Power supply → the ones in the lab for the test setup. Later we can get power supplies.~~
- Next steps:
 - Getting the parts: aluminum, brackets and nuts, another propeller
 - Finalize the CAD and start fabrication

Manav:

- Shear force should also be measured. You can have two scales or do normal and tangent; separately.
- Nuts and bolts + Aluminum profiles if needed.
- F/T sensor → ask Amin
- XY stage → ask Amin

09/12/2025 🐰

Next goal: feasibility test for each approach

Manav:

Concern: attach/detach the suction cups

- Design the robot to follow the desirable profile
- Slightly lower at the new attaching point

Testbed design:

- Test the suction cup force with the Scale (Ask Amin & Chaerim)
- Manual push and pull while recording the scale data
- Getting deformation data will be helpful
- No need for the motor system in the test setup

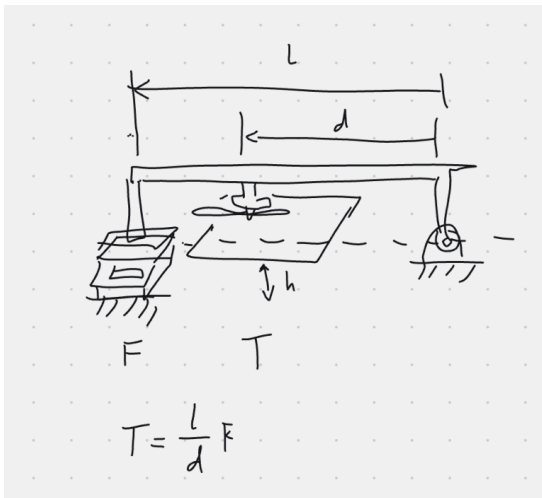
Leah:

Concern: Flight time

- Power efficiency concern
- Ceiling effect might help?

Testbed design:

- Buy propellers; NO PRINT (Justin will bring some)
- Test bed with a scale as follows:



Whiteboard link: https://zoom.us/jb/doc/Y9ZCUG63RkaQ_x15qbLk7w/p/238747068923904

09/05/2025

Milestones:

1. Problem Statement
2. Paper review and presentation
3. Required specifications:
 - a. Propeller
 - b. Suction cup: the number of the suction cups, per-cup strength, the force the cup generates, the roughness of the target surface, and the cup's softness/stiffness.
4. Simple conceptual design:
 - a. A general hand-drawing to show where to put the sensors and mechanical components.
 - b. Add annotations

09/05/2025

First week:

1. Read the selected paper (you can find it in the next pages).
 - a. Make a short presentation slide.
 - b. In what scenarios do the mechanisms fail
 - c. How do we set up the experiment test for the force data
2. System requirements: the capability to maintain surface contact and interaction force
 - a. Simple experiment for the force feasibility check
3. Locomotion method exploration
 - a. Manav: suction cup
 - i. additional consideration required to maintain surface contact (passive alignment)
 - b. Leah: quadrotor
4. Survey for components
 - a. Manav: search for suction cup models
 - b. Leah: [reference page](#) for the motors (Vertiq 23-06 2200Kv)
5. Literature list organization: see [Gelbelt lit review spreadsheet](#)
6. Communication: Through a Slack channel in the U of I robotics workspace
 - a. Chaerim will invite all and create a private channel
7. Talk with Amin/Chaerim to check out the sensor and discuss further if you have any questions.
 - a. Manav meets Amin to learn about the sensor system in person
 - b. Leah meets Chaerim to see the actuators

08/18/2025

System Requirements

Surfaces to handle

- Angles could be inclined, vertical, or even inverted surfaces.
- Surfaces may be aluminum, composite, or other materials → no magnets.
- Targeting clean surfaces to start with (dust could impact sensing performance).
- Should not damage surface (no spines, etc.).
- Overcome bumps smaller than a couple of millimeters.
- Bumps of a few millimeters just detach and come back on the other side.
- Large diameter curvature (1 m or more is probably ok), could go to larger diameters if small ones are too hard.
- Panels from Boeing for the fuselage/hull
- Energy services (pipes), nuclear facilities, airplane manufacturers (hulls and wings), and jet engine turbine blades may be tough to automate.

Sensor needs

- Apply even pressure onto the sensor surface - 10N approximately for about a 4 cm x 4 cm sensing area, 4 cm x 9 cm contact patch.
- Should not apply torsion, shear, etc. to the sensor.
- Steer by lifting the sensor
- Should not apply too much vibration (might be fine)

Potential System Concepts

1. Multirotor ceiling effect:
<https://ris.bme.cityu.edu.hk/wp-content/papercite-data/pdf/hsiao2018ceiling.pdf>
2. Suction cups: (only smooth and clean)
Two treads I found on a quick search:
<https://ieeexplore.ieee.org/abstract/document/5723554>,
<https://link.springer.com/article/10.1023/A:1022383216233>
3. Recent RA-L inchworm suction cup climber (I've seen others too):
<https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11027566>
4. Gecko adhesives look too expensive: <https://www.geckomaterials.com/purchase>
5. Magnets, spines, electrostatic adhesion, and grasping grippers seem less feasible or suited

Other Discussion Items

How many students?: about two (redundancy if one does not stay)

Timeline is flexible

Desired outputs: aim for a paper and a proposal (talk to potential funders sometime)

Wenzhen has a wet lab in DCL that folks in other labs sometimes use. Amin is the space manager. Could use for more involved fabrication.

Scheduling

- First two weeks, come up with a milestone for the students to work towards as a tryout
 - One student on each solution (multirotor and suction cups)
 - Propeller on motor tested to calibrate force for multirotor student
 - CAD design for a suction cup walker for a suction cup student.
- Create a schedule of milestones before Justin leaves for the semester.

Needed:

Range of the force to have a constant reading (at least $\sim 10\text{N}$)

Document List

1. [Literature Review Spreadsheet](#)
 - a. Keywords: the keywords used to find the literature
 - b. Title, publisher, year, link: literature info